

0 Seed Memory Program

Instructions Prompt constants for the task agent

```
1 INSTRUCTION_KNOWLEDGE_ITEM = "..."  
2 INSTRUCTION_QUERY = "..."  
3 INSTRUCTION_RESPONSE = "..."  
4
```

Schema Dataclass definitions for storage and retrieval

```
5 class KnowledgeItem: # what to store  
6     topic: str  
7     content: str  
8  
9 class Query: # how to retrieve  
10     question: str  
11     category: str  
12
```

Logic KnowledgeBase class with toolkit access

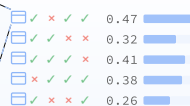
```
13 class KnowledgeBase:  
14  
15     def write(self, item: KnowledgeItem):  
16         self.db.execute(...)  
17  
18     def read(self, q: Query) → str:  
19         return self.chroma.query(...)  
20
```

5 Task-Optimized Memory



1 Program Pool

write()/read()
Instructions
Toolkit



Add to pool ▲

4 Quality Checks

Exec Limits

Output ≤3000 chars, ≤60s

▲ pass

Smoke Test

write()/read() cycle

▲ pass

Compile

Valid syntax & imports

LLM Auto-Fix

Repair errors, up to k×3

fail

Child program ▲

3 Reflect & Mutate

Diff patch
Analysis
Change log

Reflector
Diagnose eval errors
and generate patch

Current Program

✓ Success cases
✗ Failed cases
🔗 Lineage log

▼ Feedbacks

2 Evaluate

